

1 RPM Package Management

- Identify the RPM package format and explain its components.
- Use `rpm -q` to check if nano is installed.
- Query detailed package information for bash using `rpm -qi`.
- Verify the integrity of an installed package with `rpm -V`.

2 Installing & Removing RPM Packages

- Install an RPM package from a local file.
- Remove an installed package using `rpm -e`. What happens if dependencies exist?
- Use `rpm -U` to upgrade an existing package.

3 Managing Packages with YUM/DNF

- Use `yum/dnf list installed` to display installed packages.
- Search for a package using `yum/dnf search`.
- Install `htop` using `yum/dnf install`.
- Remove `htop` with `yum/dnf remove`.

4 Configuring YUM/DNF Repositories

- List enabled repositories using `yum/dnf repolist`.
- Add a new repository by creating a `.repo` file in `/etc/yum.repos.d/`.
- Disable all repositories except `cdrom.repo` and verify the changes.

Week 13 practical – Using a local CentOS DVD repository

The slides provide a procedure for configuring YUM or DNF to use a local or remote ISO repository. They instruct the user to insert the CentOS 9 DVD, inspect the optical drive mount, create `cdrom.repo`, disable other repositories and confirm the enabled repository list.

The exact `baseurl` depends on where the DVD is mounted on your VM. Do not copy an assumed path without checking it.

Step 1: Insert the installation ISO

In the VM settings, attach the CentOS installation ISO to the optical drive.

Then check whether it is mounted:

```
mount | grep sr0
```

You may see output similar to:

```
/dev/sr0 on /run/media/root/CentOS-Stream-9-BaseOS-x86_64 type iso9660 ...
```

Record the actual mount path shown after `on`.

Also check:

lsblk -f

Step 2: Check the DVD contents

Replace the path below with your actual mount location:

ls /run/media/root/CentOS-Stream-9-BaseOS-x86_64

The repository may contain directories such as:

BaseOS

AppStream

Check for repository metadata:

find /run/media/root/CentOS-Stream-9-BaseOS-x86_64 \

-name repodata -type d

The parent directory containing **repodata** is the directory that should normally be used as a repository **baseurl**.

Step 3: Create the repository file

sudo vi /etc/yum.repos.d/cdrom.repo

A CentOS 9 DVD commonly separates BaseOS and AppStream, so the file may require two repository sections.

Use the actual ISO mount path detected on your VM:

[cdrom-baseos]

name=CentOS DVD BaseOS

baseurl=file:///ACTUAL-MOUNT-PATH/BaseOS

enabled=1

gpgcheck=0

```
[cdrom-appstream]
```

```
name=CentOS DVD AppStream
```

```
baseurl=file:///ACTUAL-MOUNT-PATH/AppStream
```

```
enabled=1
```

```
gpgcheck=0
```

The uploaded slide's repository text is not fully preserved in the extracted content, so the exact repository IDs and paths above are a practical structure rather than a verbatim transcription.

The mount path must be based on `mount | grep sr0`.

Save and exit:

Esc

:wq

Step 4: Disable other repositories for testing

A safe one-command test is:

```
sudo dnf --disablerepo='*' \
```

```
--enablerepo=cdrom-baseos,cdrom-appstream \
```

```
repolist
```

Expected result: only the local DVD repositories appear.

For persistent disabling, first list the exact repository IDs:

```
dnf repolist --all
```

Then use:

```
sudo dnf config-manager --set-disabled REPOSITORY_ID
```

Do not disable repositories blindly without recording their names, because you may need to restore them later.

Step 5: Clean the metadata cache

```
sudo dnf clean all
```

Rebuild it using the DVD repositories:

```
sudo dnf --disablerepo='*' \  
--enablerepo=cdrom-baseos,cdrom-appstream \  
makecache
```

Step 6: Confirm the repository

```
dnf repolist
```

or:

```
yum repolist
```

The slides specifically require using `yum repolist` or `dnf repolist` to confirm the enabled repositories.

Step 7: Search for a package using the DVD

```
dnf --disablerepo='*' \  
--enablerepo=cdrom-baseos,cdrom-appstream \  
search httpd
```

Display its information:

```
dnf --disablerepo='*' \  
--enablerepo=cdrom-baseos,cdrom-appstream \  
info httpd
```

Step 8: Install a package using the DVD

```
sudo dnf --disablerepo='*' \
--enablerepo=cdrom-baseos,cdrom-appstream \
install httpd

Verify:

rpm -q httpd

List its files:

rpm -ql httpd

Check its information:

rpm -qi httpd
```

Step 9: Remove the test package

```
sudo dnf remove httpd

Verify:

rpm -q httpd

Expected:

package httpd is not installed
```

Be aware that if `httpd` was already required for another lab or project, removing it may disrupt that work.

Week 13 practical – RPM query and verification workflow

Use an available `.rpm` file from the mounted installation DVD.

Find one:

```
find /run/media -type f -name "*.rpm" | head
```

Store the selected path in a variable:

PACKAGE=/full/path/to/package.rpm

Display package information:

```
rpm -qip "$PACKAGE"
```

List its files:

```
rpm -qpl "$PACKAGE"
```

List dependencies:

```
rpm -qpR "$PACKAGE"
```

View scripts:

```
rpm -qp --scripts "$PACKAGE"
```

Import keys:

```
sudo rpm --import /etc/pki/rpm-gpg/*
```

List imported keys:

```
rpm -qa 'gpg-pubkey*'
```

Verify the package:

```
rpm -K "$PACKAGE"
```

This workflow allows the package to be inspected before it is installed.

Week 13 practical – DNF package-management workflow

Step 1: Examine repositories

```
dnf repolist
```

```
dnf repolist --all
```

Step 2: Search for a package

```
dnf search tree
```

```
dnf info tree
```

Step 3: Check whether it is installed

```
rpm -q tree
```

Step 4: Install it

```
sudo dnf install tree
```

Step 5: Verify it

```
rpm -q tree
```

```
which tree
```

```
tree --version
```

Step 6: Identify its files

```
rpm -ql tree
```

Step 7: Identify which package owns its command

```
rpm -qf "$(which tree)"
```

Step 8: Remove it

```
sudo dnf remove tree
```

Step 9: Verify removal

```
rpm -q tree
```

Common Week 13 mistakes

Using an RPM filename when removing an installed package

Incorrect:

```
sudo rpm -e httpd-2.4.57.x86_64.rpm
```

Correct:

```
sudo rpm -e httpd
```

`rpm -e` uses the installed package name.

Forgetting `-p` when querying an RPM file

Installed package:

```
rpm -qi httpd
```

RPM file:

```
rpm -qip httpd.rpm
```

Without `-p`, RPM searches the installed package database.

Installing directly with RPM despite missing dependencies

This may fail:

```
sudo rpm -ivh httpd.rpm
```

Prefer:

```
sudo dnf install ./httpd.rpm
```

when permitted, because DNF can resolve dependencies from configured repositories.

The `dnf install ./file.rpm` form is useful general DNF behaviour, but the uploaded slides demonstrate dependency handling mainly through RPM and YUM repository commands.

Using `--nodeps` to force installation

Avoid:

```
sudo rpm -ivh --nodeps package.rpm
```

This may install broken software because required dependencies are ignored.

Assuming the ISO mount path

Do not assume:

/media

or:

/run/media/root/CentOS

Check:

`mount | grep sr0`

Then use the actual path in `baseurl`.

Using the wrong number of slashes in a local repository

For a local absolute path:

`baseurl=file:///path/to/repository`

There are three slashes after `file::`

`file:///`

Pointing `baseurl` at the wrong directory

The directory should contain:

`repodata/`

Check:

`ls /repository/path/repodata`

If `repodata` is absent, DNF cannot treat it as a repository.

Forgetting to clear repository metadata

After changing repository files:

`sudo dnf clean all`

```
sudo dnf makecache
```

Disabling every repository permanently

Before disabling repositories, record their IDs:

```
dnf repolist --all
```

For testing, it is safer to use:

```
dnf --disablerepo='*' --enablerepo=REPO_ID command
```

This affects only the current command.

Week 13 essential commands

Command	Purpose
<code>rpm -q PACKAGE</code>	Check whether a package is installed
<code>rpm -qa</code>	List all installed packages
<code>rpm -qi PACKAGE</code>	Display installed package information
<code>rpm -qip FILE.rpm</code>	Display information about an RPM file
<code>rpm -ql PACKAGE</code>	List files belonging to a package
<code>rpm -qc PACKAGE</code>	List configuration files

<code>rpm -qd PACKAGE</code>	List documentation files
<code>rpm -qR PACKAGE</code>	List package requirements
<code>rpm -qf FILE</code>	Identify the package owning a file
<code>rpm -qp --scripts FILE.rpm</code>	Inspect RPM scripts
<code>rpm -K FILE.rpm</code>	Verify an RPM file
<code>rpm -V PACKAGE</code>	Verify installed package files
<code>rpm -ivh FILE.rpm</code>	Install an RPM file
<code>rpm -Uvh FILE.rpm</code>	Upgrade or install an RPM
<code>rpm -e PACKAGE</code>	Remove an installed package
<code>dnf repolist</code>	List enabled repositories
<code>dnf search TERM</code>	Search for packages
<code>dnf info PACKAGE</code>	Display package information

`dnf provides PATH` Find the package providing a file

`dnf install PACKAGE` Install with dependencies

`dnf remove PACKAGE` Remove a package

`dnf check-update` Check for updates

`dnf update` Update packages

`dnf group list` List package groups

`dnf group info` Display group information

`dnf group install` Install a package group

`dnf history` Display package transactions